Henry Routson

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Embedded AI engineer shipping production-grade GenAI: agentic workflows, typed data pipelines, evaluation, and observability. Master's in AI (in progress).

Education

The University of Melbourne
Master of Computer Science, AI — *in progress, part time*

- NLP: Fine Tuning, Prompt Tuning, Transformers, Hugging Face

The University of Melbourne
Bachelor of Science, Computing and Software Systems — *Distinction*

Ballarat Grammar — Academic scholarship; VCE Software Development, Specialist & Methods Mathematics

Achievements

Antler VC Build Hackathon, 1st Place Jun 2026
antler.co | Event page

- Won 1st place and $7,500 at Antler's two-day Sydney sprint.
- 14-hour prototype of Spine (auto-pipeline.dev / app).

Experience

KPMG, AI Engineer Jul 2025 – Present
kpmg.com/au/en.html

- Embedded AI Prototype Engineer in Futures: technical discovery through production-grade LLM and agent systems.
- Shipped a 34k-line Python platform for prompting, structured parsing, Pydantic validation, and deep research.
- Built pipelines over unstructured and structured data: 10k-page legislative assessment; SQL migration and database merge.
- Multi-agent and hierarchical sub-agent systems with Recursive Language Models (RLM); deployed Transformers on CUDA (Hugging Face, Ollama).
- *Skills:* Python, Azure, FastAPI, Pydantic, LangChain, LangExtract, Hugging Face, Docker, Ray, Pandas, DeepSeek-OCR, Ollama.

Spine, Founder (Antler AS16) Jul 2026 – Present
spine-ai.dev | antler.co/australia

- 80 of 1,600 selected for Antler Australia's inception-stage VC residency; shipping Spine as founder-engineer.
- Production GenAI pipeline platform: document ingestion to typed structures (Pydantic/Zod/JSON Schema) to parallel LLM-batch to validated structured output, with step caching.
- Four-surface parity: GUI, CLI/TUI, HTTP APIs, and MCP servers sharing one domain model (projects, pipelines, runs, traces).
- Financial-services templates: bank reconciliation, expense-policy audit, tax workpapers, procurement three-way match, audit sampling, KYC/compliance, Payment Times Reporting (PTR).
- Evaluation and observability: per-step tests (deterministic vs LLM-as-judge), granular tracing (tokens/sec, cost-per-request, latency), prompt profiling, Langfuse export, human-in-the-loop.
- RAG-like quote grounding with source offsets; OCR/PDF/Excel/Word; custom LLM providers; Docker/Supabase deploys; code export (Python, TypeScript, Rust).
- *Skills:* Next.js, TypeScript, Python, Zod, Supabase, Postgres, Docker, OpenRouter (Gemini, DeepSeek), MCP, Langfuse, Stripe.

Bet Right, Data Science Intern Jan 2025 – Apr 2025
linkedin/bet-right

- Financial-services data stack: SQL, Snowflake, Databricks, PowerBI, Azure, DevOps.
- Improved abuse-detection ML model by 15–40% using stacking.

DSCubed, DS/AI Team Vice Head Nov 2024 – Jul 2025
dscubed.org.au

- RAG, vector databases, and LLM APIs for student-facing AI applications.

ARES, TypeScript Developer, University of Melbourne 2023
linkedin/aresunimelb

- 3D front-end for rocket trajectory simulation (supersonic and subsonic).

Technical Skills

GenAI / agents: multi-agent systems, MCP servers, tool calling, RAG, vector databases, structured extraction, Pydantic, Zod, JSON Schema, schema validation, human-in-the-loop, state management, typed pipelines
Eval / observability: evaluation pipelines, granular tracing, tokens/sec, cost-per-request, latency, Langfuse, prompt profiling, A/B testing, LLM-as-judge
Data / cloud: Python, TypeScript, SQL, Rust, FastAPI, Azure, Docker, Postgres, Snowflake, Databricks, OCR, APIs, CI
Financial services AI: KYC, bank reconciliation, expense audit, tax workpapers, procurement match, PTR, compliance pipelines

Certifications

Microsoft Certified: Azure Fundamentals Apr 2025
learn.microsoft.com

Open-Source

Cached File Explorer (Rust) — PR #47: poisoned-thread fix; caching / low-level Rust.

Supabase SQL Auth UI — PR #108: options and docs.

Projects

LLM Experiments github/llm_experiments

- Transformers, JAX, Keras, PyTorch; pre-training plus supervised fine-tuning.

Auto Avail (TypeScript) github/Auto_Avail

- Location-aware scheduler. Supabase, Flutter, React Native, Postgres, pgTAP, Expo.

LeetCode: leetcode.com/u/HenryRoutson

References

- **Cindy** — AI Engineer, KPMG. LinkedIn
- **Pranav** — L3 Data Scientist, NAB. LinkedIn